Lecture 10: Neural Machine Translation and Models with Attention

Notebook:	cs224N Natural Language Processing		
Created:	2019/6/9 13:39	**Updated:**	2019/6/9 14:31
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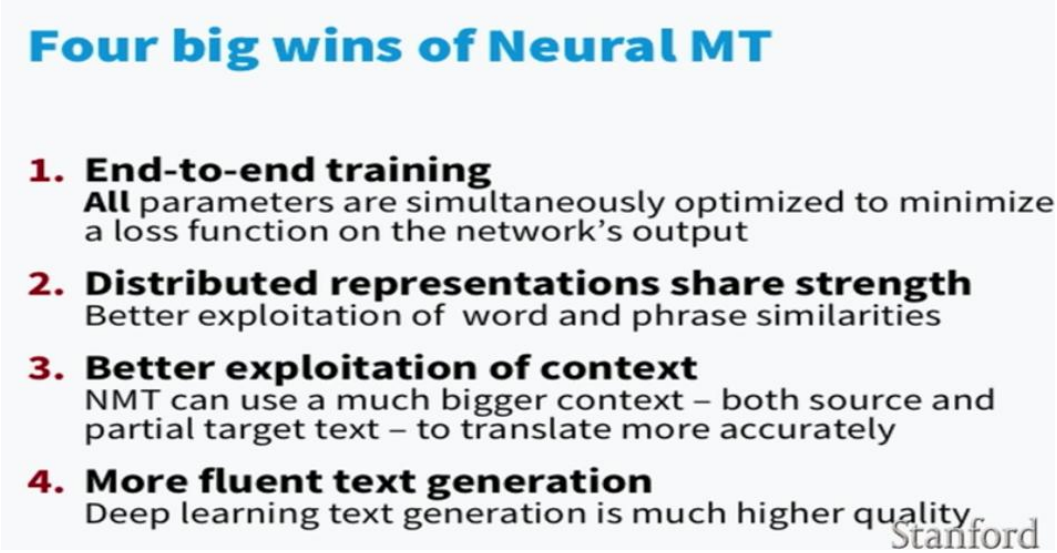
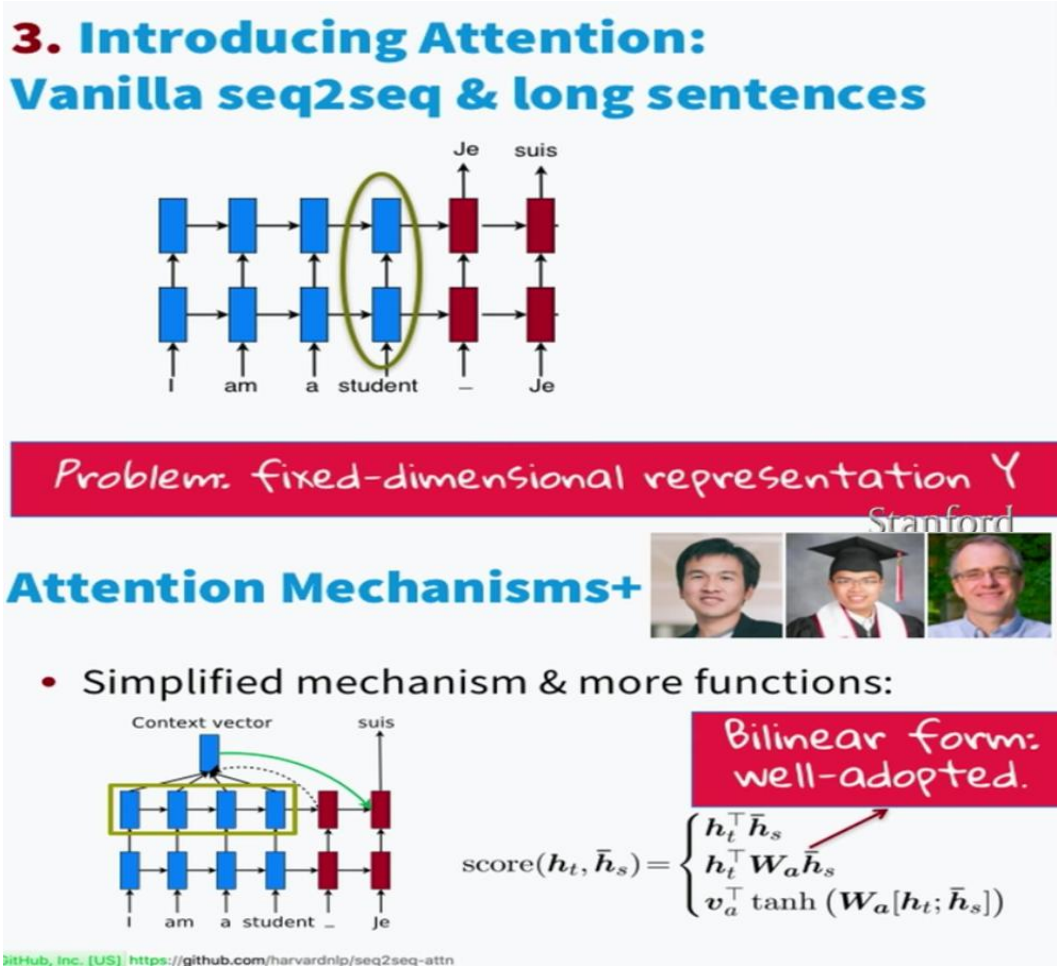
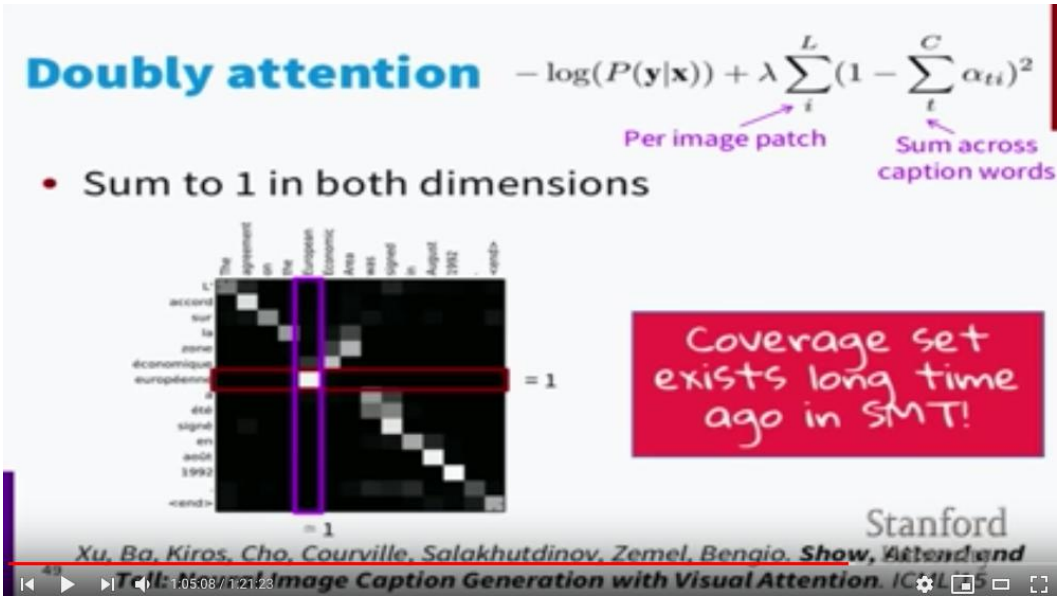


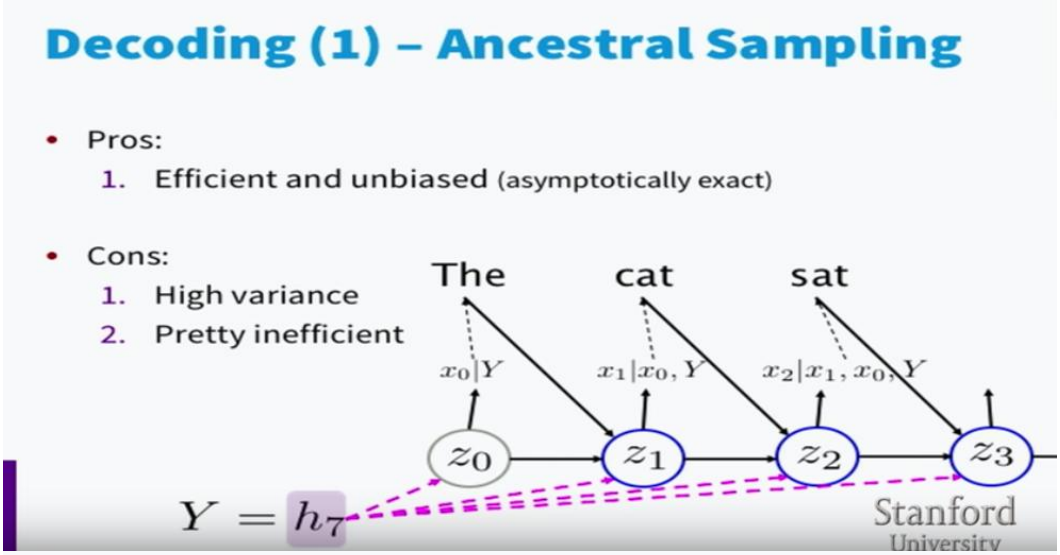
Attention: MT ex.

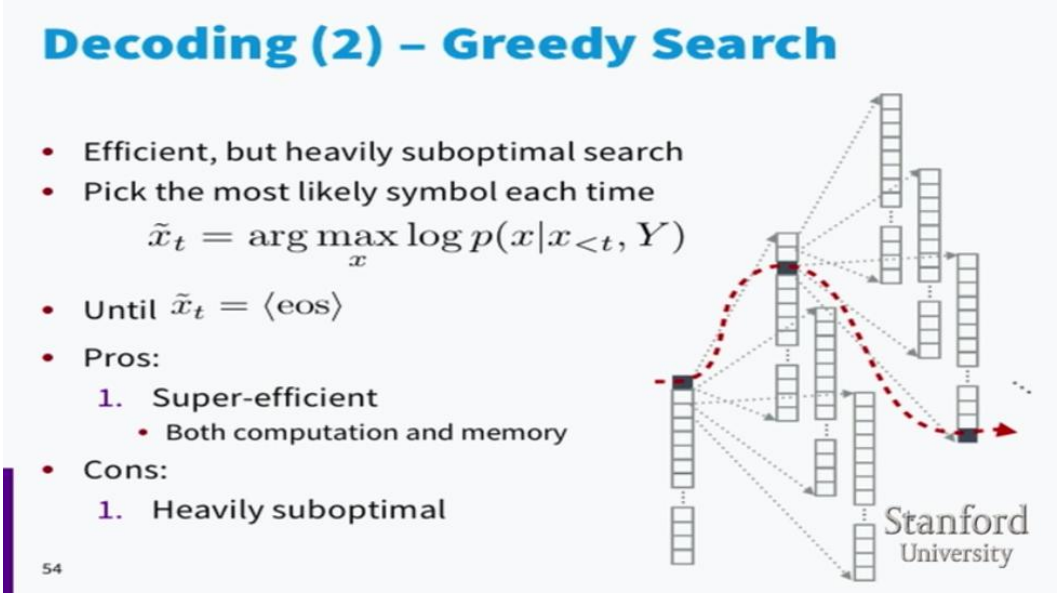




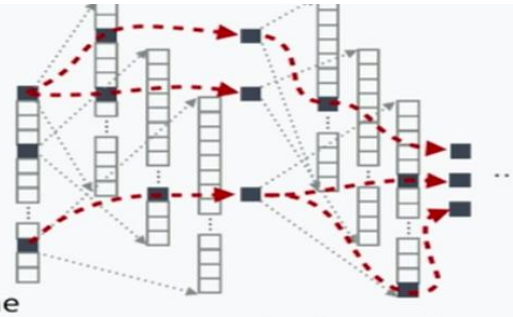
Coverage-based attention: image caption

decode





Decoding (3) – Beam Search



- Pretty, but *not very* efficient
- Maintain *K* hypotheses at a time

$$\mathcal{H}_{t-1} = \{(\tilde{x}_1^1, \tilde{x}_2^1, \ldots, \tilde{x}_{t-1}^1), (\tilde{x}_1^2, \tilde{x}_2^2, \ldots, \tilde{x}_{t-1}^2), \ldots, (\tilde{x}_1^K, \tilde{x}_2^K, \ldots, \tilde{x}_{t-1}^K)\}$$

- Expand each hypothesis

$$\mathcal{H}_t^k = \{(\tilde{x}_1^k, \tilde{x}_2^k, \ldots, \tilde{x}_{t-1}^k, v_1), (\tilde{x}_1^k, \tilde{x}_2^k, \ldots, \tilde{x}_{t-1}^k, v_2), \ldots, (\tilde{x}_1^k, \tilde{x}_2^k, \ldots, \tilde{x}_{t-1}^k, v_{|V|})\}$$

- Pick top-K hypotheses from the union $\mathcal{H}_t = \cup_{k=1}^K \mathcal{B}_k$, where

$$\mathcal{B}_k = \arg\max_{\tilde{X} \in \mathcal{A}_k} \log p(\tilde{X}|Y), \ \mathcal{A}_k = \mathcal{A}_{k-1} - \mathcal{B}_{k-1}, \text{ and } \mathcal{A}_1 = \cup_{k'=1}^K \mathcal{H}_t^{k'}.$$

Stanford